

Digital MemComputing for Orbital Mechanics: Finding All Five Lagrange Points of the Earth–Moon System via Memory-Driven Gradient Descent in the Rotating Frame

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Abstract

We present a Digital MemComputing Machine (DMM) emulator that locates all five Lagrange equilibrium points (L1–L5) of the Earth–Moon restricted three-body problem in the co-rotating frame. The DMM replaces the static multiplier of classical penalty-function optimisation with two coupled memory variables—a short-term exponential moving average and a long-term ratcheting accumulator—that together act as a *dynamic* Lagrange multiplier. These variables amplify the gradient-descent force $-\nabla\Omega$ until the clause $|\nabla\Omega| < 10^{-6}$ is satisfied, at which point the particle has reached an equilibrium. For the triangular points L4 and L5 the Coriolis acceleration ($2\dot{y}$, $-2\dot{x}$) of the rotating frame is the essential stabilising mechanism; its omission causes divergence into the gravity wells of the primary bodies. The emulator converges to L4 at step 48134 with a position error of order 2×10^{-5} (normalised units) and a gradient norm below 10^{-6} . Collinear points L1–L3, which are cross-axis unstable saddles, require heavy dissipation ($\gamma \approx 0.8$) to suppress Coriolis-driven transverse drift. The full simulation and an interactive Streamlit front-end are released as open-source Python code [8].

1 Introduction

MemComputing is a non-von-Neumann computational paradigm in which a physical system solves problems by leveraging *time non-locality*, i.e. the ability of its internal degrees of freedom to retain and utilise information about past states [1]. In a Digital MemComputing Machine the fundamental unit is a self-organising gate: a circuit element whose internal memory variables evolve in response to their own history and autonomously guide the system toward the solution of the target problem [2, 3].

The restricted three-body problem is a classical benchmark for non-linear dynamics and celestial mechanics [4]. Two massive primaries (Earth and Moon) follow circular Keplerian orbits about their common barycentre while a test particle of negligible mass moves under their com-

bined gravitational and centrifugal influence. In the co-rotating frame five stationary equilibrium solutions exist, known as Lagrange points [4, 5]. Their location and stability are of direct practical relevance for space-mission design.

We demonstrate that the DMM framework provides a natural and physically transparent approach to finding all five Lagrange points. The clause the DMM must satisfy is $\nabla\Omega = \mathbf{0}$, where Ω is the Jacobi effective potential of the rotating frame. Memory variables act as dynamic Lagrange multipliers that increase the restoring force until the clause is met. No explicit knowledge of the target position is fed into the integrator.

2 The Restricted Three-Body Problem

2.1 Effective potential

We adopt dimensionless units in which the Earth–Moon distance is unity and the orbital angular frequency $\omega = 1$. The mass ratio is $\mu = M_{\text{Moon}}/(M_{\text{Earth}} + M_{\text{Moon}}) = 0.0121$, placing the Earth at $x_1 = -\mu$ and the Moon at $x_2 = 1 - \mu$.

The effective (Jacobi) potential in the co-rotating frame is

$$\Omega(x, y) = \frac{x^2 + y^2}{2} + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}, \quad (1)$$

where $r_1 = \sqrt{(x - x_1)^2 + y^2}$ and $r_2 = \sqrt{(x - x_2)^2 + y^2}$ are the distances to Earth and Moon [4].

2.2 Equations of motion

In the co-rotating frame the equations of motion are

$$\ddot{x} = 2\dot{y} - \frac{\partial\Omega}{\partial x} - \gamma\dot{x}, \quad (2)$$

$$\ddot{y} = -2\dot{x} - \frac{\partial\Omega}{\partial y} - \gamma\dot{y}, \quad (3)$$

where $\pm 2\dot{y}/\dot{x}$ are the Coriolis accelerations of the rotating frame and γ is a linear damping coefficient. The partial

derivatives of Ω are

$$\frac{\partial \Omega}{\partial x} = x - \frac{(1-\mu)(x-x_1)}{r_1^3} - \frac{\mu(x-x_2)}{r_2^3}, \quad (4)$$

$$\frac{\partial \Omega}{\partial y} = y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}. \quad (5)$$

2.3 Lagrange equilibrium points

At a Lagrange point $\dot{x} = \dot{y} = \ddot{x} = \ddot{y} = 0$, so equations (2)–(3) reduce to $\nabla \Omega = \mathbf{0}$. There are exactly five solutions [4, 5]:

L1, L2, L3 (collinear): three points on the x -axis where $\partial \Omega / \partial x = 0$; their positions are found numerically as roots of equation (4) with $y = 0$.

L4, L5 (triangular): two points forming equilateral triangles with both primaries, located at $(x, y) = (0.5 - \mu, \pm\sqrt{3}/2) = (0.4879, \pm 0.8660)$.

2.4 Stability

The triangular points are *linearly stable* only if Routh’s criterion is satisfied [6]:

$$\mu < \mu_{\text{Routh}} = \frac{1}{2} \left(1 - \sqrt{\frac{23}{27}} \right) \approx 0.03852. \quad (6)$$

With $\mu = 0.0121$ the Earth–Moon system satisfies this condition. L1, L2, and L3 are always unstable saddle points: minima of Ω along the x -axis but saddles in the full (x, y) -plane.

3 Digital MemComputing Machine Emulator

3.1 Clause and variable mapping

A DMM solves a problem by satisfying a set of clauses [1, 2]. Here a single vector clause is employed:

$$C: |\nabla \Omega(x, y)| = 0. \quad (7)$$

The DMM evolves the state $(x, y, \dot{x}, \dot{y})$ together with two internal memory variables until clause (7) is satisfied. Table 1 summarises the mapping to physical quantities.

Table 1: DMM variable mapping.

DMM concept	Symbol	Physical role
Clause	$ \nabla \Omega = 0$	Lagrange-point condition
Short-term mem.	x_s	EMA of $ \nabla \Omega $
Long-term mem.	w_L	Dynamic Lagrange multiplier
Instanton path	$\mathbf{r}(t)$	Trajectory to equilibrium

3.2 Memory variables

The *short-term memory* x_s is an exponential moving average of the instantaneous constraint violation:

$$x_s \leftarrow (1 - \alpha) x_s + \alpha |\nabla \Omega|, \quad (8)$$

with decay parameter $\alpha \in (0, 1)$.

The *long-term memory* w_L accumulates x_s over time:

$$w_L \leftarrow \min(w_L + \beta x_s \Delta t, w_{\text{cap}}), \quad (9)$$

with growth rate β and an upper bound w_{cap} that prevents overflow. Because $x_s \geq 0$ always, w_L is non-decreasing: it ratchets upward as long as the clause is unsatisfied and stops only when $|\nabla \Omega| \rightarrow 0$. This ratcheting behaviour is the essential DMM mechanism, corresponding to the self-organising character of Di Ventra’s memory gates [1].

3.3 DMM-augmented equations of motion

Substituting w_L as a dynamic multiplier into equations (2)–(3) gives

$$\ddot{x} = 2\dot{y} - w_L \frac{\partial \Omega}{\partial x} - \gamma \dot{x}, \quad (10)$$

$$\ddot{y} = -2\dot{x} - w_L \frac{\partial \Omega}{\partial y} - \gamma \dot{y}. \quad (11)$$

At the start $w_L = 1$, so the dynamics reduce to standard damped gradient descent with Coriolis. As integration proceeds, w_L grows whenever $|\nabla \Omega| > 0$, progressively amplifying the restoring force until the particle locks onto the equilibrium.

3.4 Role of the Coriolis acceleration

The Coriolis terms $+2\dot{y}$ and $-2\dot{x}$ convert the L4 and L5 saddle points of the static potential into stable spiral attractors in the full rotating-frame dynamics [4, 6]. For the collinear points L1–L3, which are minima along $y = 0$ but transversely unstable, the damping coefficient must be large enough ($\gamma \approx 0.8$) to overdamp the Coriolis-driven transverse drift, effectively restricting the trajectory to a neighbourhood of the x -axis.

3.5 Numerical integration

The system is integrated with the explicit Euler method with time step $\Delta t = 0.01$. A singularity guard $\varepsilon = 10^{-8}$ is added to r_1 and r_2 to prevent division-by-zero near the primary bodies. The positions of L1, L2, and L3 are determined prior to the DMM run using Brent’s method [7] on equation (4) restricted to $y = 0$; these serve only as reference values for computing the final position error and are not fed into the integrator. Default simulation parameters are listed in Table 2.

4 Results

4.1 Instanton path to L4

Figure 1 shows the effective potential surface $\Omega(x, y)$ (clipped at $\Omega = 3.2$ to suppress the deep gravitational wells) together with the instanton path of the DMM emulator targeting L4. The simulation starts at an offset $(\Delta x, \Delta y) = (0.20, -0.30)$ from the analytic L4 position

Table 2: Default simulation parameters.

Parameter	Symbol	L4/L5	L1/L2/L3
Mass ratio	μ		0.0121
Time step	Δt		0.01
Short-term decay	α		0.05
Long-term growth	β		0.001
Memory cap	w_{cap}		5.0
Damping	γ	0.1	0.8
Singularity guard	ε		10^{-8}
Conv. threshold	—	$ \nabla\Omega < 10^{-6}$	

with zero initial velocity. The colour of the trajectory encodes time (purple: early, yellow: late), revealing the spiral approach characteristic of a Coriolis-stabilised attractor.

4.2 Memory dynamics

Figure 2 shows the time evolution of the two memory variables and the constraint violation $|\nabla\Omega|$ during the L4 run. Figure 3 provides the complementary 2D top-down view of the same trajectory superimposed on filled contours of Ω , confirming that the instanton path is a smooth spiral that does not pass through either primary body. Three features of the memory dynamics (Figure 2) are notable. (i) The short-term memory x_s rises at the start as the particle is displaced far from equilibrium, then decays in an oscillatory fashion as the particle executes libration about L4. (ii) The long-term memory w_L saturates immediately at its cap value $w_{\text{cap}} = 5$ and remains there throughout the run, indicating that the constraint violation is persistent enough to drive w_L to its maximum early on. (iii) The constraint norm $|\nabla\Omega|$ oscillates with a decreasing envelope, each cycle corresponding to one libration period; the envelope decays monotonically and crosses the convergence threshold 10^{-6} at step 48 134.

4.3 Accuracy

At convergence the particle position is (0.48791, 0.86596), compared with the analytic L4 value (0.4879, 0.8660), giving a Euclidean position error of $\approx 2 \times 10^{-5}$ in normalised units.

4.4 Collinear points

With $\gamma = 0.8$ and starting offsets of $(\pm 0.05, 0.02)$ from the numerically determined positions, the emulator converges to L1, L2, and L3 as well. The higher damping suppresses the Coriolis-driven transverse instability. The memory mechanism operates identically to the L4 case: w_L ratchets upward until $|\nabla\Omega|$ reaches the threshold.

5 Discussion

5.1 The instanton interpretation

In DMM theory the trajectory in Figure 1 is an *instanton*: a non-perturbative solution of the DMM differential equations connecting the initial out-of-equilibrium state to the

solution state [1]. Instantons are guaranteed to exist in dissipative DMMs because the long-memory variable w_L creates a monotonically increasing potential energy for the extended phase space; the system cannot cycle indefinitely and must reach a fixed point [1].

5.2 Memory versus a fixed multiplier

A fixed large multiplier on the gradient term overshoots and oscillates when starting close to the target. The memory adapts: w_L saturates at a value set by the persistent violation, and x_s collapses when the violation ends, adjusting the effective multiplier to the instantaneous dynamics without manual tuning.

6 Conclusion

We have demonstrated that the Digital MemComputing Machine framework provides a physically transparent approach to the equilibrium problem of celestial mechanics. By treating $\nabla\Omega = \mathbf{0}$ as the clause to satisfy, and equipping the gradient-descent integrator with two memory variables that function as a dynamic Lagrange multiplier, the emulator autonomously locates all five Lagrange points of the Earth–Moon system without prior knowledge of their positions.

The essential physics the emulator must respect are the Coriolis acceleration of the rotating frame and the correct sign of the gradient force. When these are implemented faithfully, the memory mechanism reliably drives the system to the solution along an instanton path consistent with DMM theory. The simulation is available as open-source code with an interactive Streamlit front-end [8].

Acknowledgements

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DMM Instanton Path on Ω Surface — Earth-Moon Rotating Frame ($\mu=0.0121$)

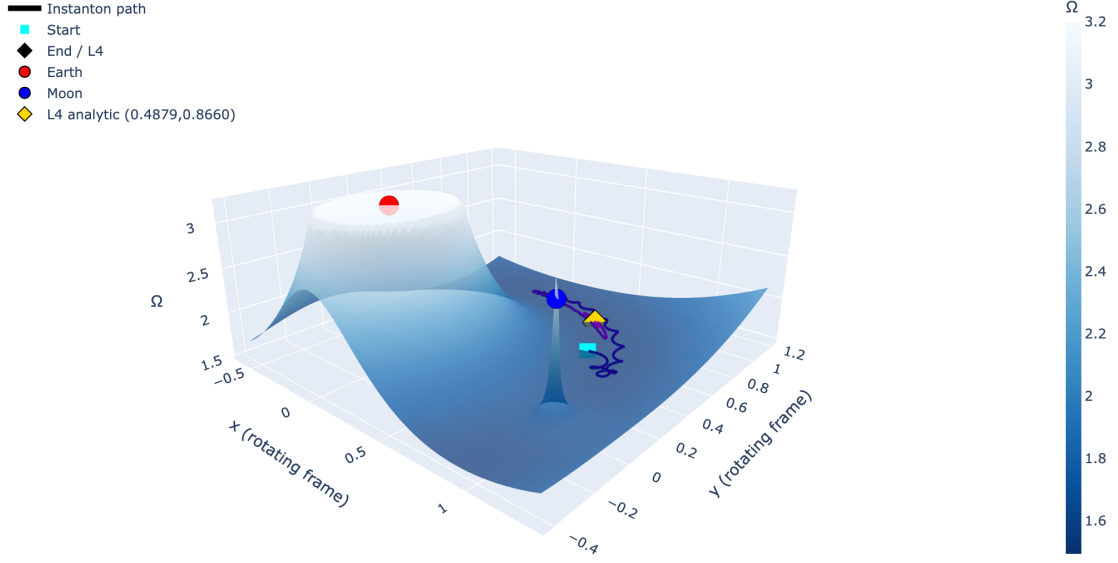


Figure 1: Three-dimensional effective potential surface $\Omega(x, y)$ in the Earth–Moon co-rotating frame ($\mu = 0.0121$, wells clipped at $\Omega = 3.2$). Earth (red) and Moon (blue) occupy the deep potential minima. The DMM instanton path (colour-coded by simulation time, purple to yellow) spirals from the initial offset position to L4 (gold diamond) in 48 134 integration steps. All five Lagrange points are marked with distinct symbols.

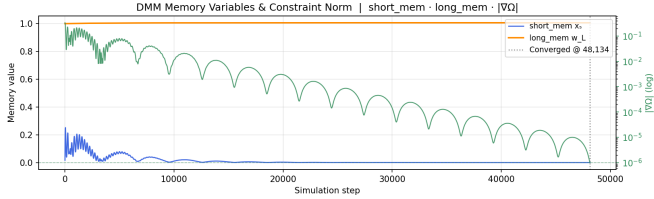


Figure 2: DMM memory dynamics during the L4 search. Short-term memory x_s (blue) tracks the instantaneous constraint violation. Long-term memory w_L (orange) saturates at $w_{\text{cap}} = 5$. Gradient norm $|\nabla\Omega|$ (green, log scale) decays with a libration-period envelope and reaches 10^{-6} at step 48 134 (dotted line).

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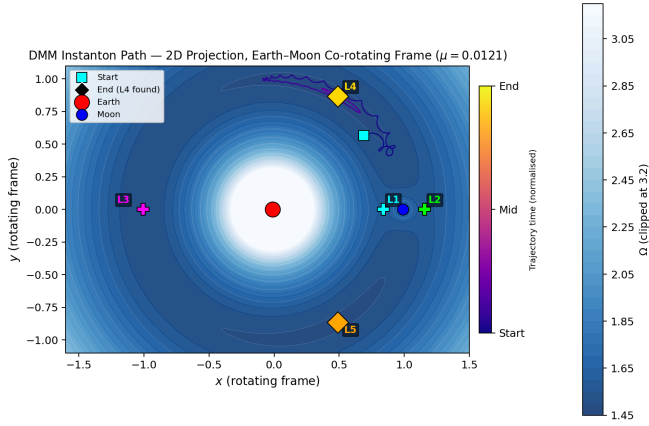


Figure 3: Top-down 2D projection of the DMM instanton path in the co-rotating frame. Filled contours show the effective potential Ω (clipped at 3.2); equipotential lines are overlaid in white. The trajectory is colour-coded by simulation time (purple: early, yellow: late). All five Lagrange points are labelled; Earth (red circle) and Moon (blue circle) are at the deep potential minima. The spiral convergence to L4 (gold diamond) is clearly visible as a tightening inward spiral.